

JAVA DEVELOPMENT

|

| — 1. CORE JAVA — FOUNDATION

| |

| | — 1.1 Java Basics

| | | — JDK, JRE, JVM

| | | — Variables

| | | — Data Types

| | | — Type Casting

| | | — Operators

| | | — Input / Output

| | | — if-else

| | | — switch

| | | — for / while / do-while

| | | — Arrays

| | | — Methods

| |

| | — 1.2 OOP

| | | — Classes & Objects

| | | — Constructors

| | | — this / super

| | | — Encapsulation

| | | — Inheritance

| | | — Polymorphism

| | | — Abstraction

| | | — Interfaces

| | | — static vs instance

| | | — Access Modifiers

| |

| | — 1.3 Core Concepts

- | | ├── String
- | | ├── StringBuilder
- | | ├── StringBuffer
- | | ├── Wrapper Classes
- | | ├── Autoboxing / Unboxing
- | | ├── final / finally
- | | ├── Packages
- | | └── Java Memory Basics
- | |
- | ├── 1.4 Exception Handling
- | | ├── try / catch / finally
- | | ├── throw / throws
- | | ├── Checked Exceptions
- | | ├── Unchecked Exceptions
- | | └── Custom Exceptions
- | |
- | ├── 1.5 Collections
- | | ├── List
- | | | ├── ArrayList
- | | | └── LinkedList
- | | ├── Set
- | | | ├── HashSet
- | | | ├── LinkedHashSet
- | | | └── TreeSet
- | | ├── Map
- | | | ├── HashMap
- | | | ├── LinkedHashMap
- | | | └── TreeMap
- | | ├── Queue
- | | └── Deque

- | | | — PriorityQueue
- | | | — Iterator
- | | | — Comparable / Comparator
- | |
- | | — 1.6 Generics
- | | | — Generic Classes
- | | | — Generic Methods
- | | | — Bounded Types
- | | | — Wildcards
- | |
- | | — 1.7 Modern Java
- | | | — Lambda Expressions
- | | | — Functional Interfaces
- | | | — Predicate
- | | | — Consumer
- | | | — Supplier
- | | | — Function
- | | | — Method References
- | | | — Stream API
- | | | — Optional
- | | | — var
- | | | — Records
- | | | — Sealed Classes
- | | | — Pattern Matching
- | |
- | | — 1.8 File I/O
- | | | — File
- | | | — FileReader / Writer
- | | | — BufferedReader / Writer
- | | | — Serialization

- | | └─ NIO / Path / Files
- | |
- | └─ 1.9 Multithreading & Concurrency ★
- | └─ Thread
- | └─ Runnable
- | └─ Callable
- | └─ Thread Lifecycle
- | └─ synchronized
- | └─ wait / notify / notifyAll
- | └─ ExecutorService
- | └─ Thread Pools
- | └─ Concurrent Collections
- | └─ ConcurrentHashMap
- | └─ Locks
- | └─ Race Conditions
- | └─ Deadlocks
- | └─ CompletableFuture
- |
- | └─ 2. DEVELOPER TOOLS
- | |
- | └─ IntelliJ IDEA
- | └─ Debugging
- | └─ Breakpoints
- | └─ Git
- | └─ GitHub
- | └─ Branching
- | └─ Merge / Rebase
- | └─ Pull Requests
- | └─ Conflict Resolution
- | └─ Maven

- | | | — pom.xml
- | | | — Dependencies
- | | | — Lifecycle
- | | — Gradle Basics
- |

| — 3. WEB FUNDAMENTALS

- | |
- | | — Client / Server
- | | — HTTP / HTTPS
- | | — HTTP Methods
- | | | — GET
- | | | — POST
- | | | — PUT
- | | | — PATCH
- | | | — DELETE
- | | — HTTP Status Codes
- | | — Headers
- | | — Request / Response
- | | — JSON
- | | — REST API
- | | — REST Principles
- | | — API Versioning
- | | — Postman
- |

| — 4. DATABASE + SQL

- | |
- | | — MySQL / PostgreSQL
- | | — Database & Tables
- | | — CRUD
- | | — SELECT / WHERE

- | |— ORDER BY
- | |— GROUP BY
- | |— HAVING
- | |— JOINS
- | |— Subqueries
- | |— Constraints
- | |— Primary / Foreign Keys
- | |— Normalization
- | |— Indexes
- | |— Transactions
- | |— ACID
- | |— JDBC
 - | |— Connection
 - | |— Statement
 - | |— PreparedStatement
 - | |— ResultSet
 - | |— CRUD
 - | |— Transactions
 - | |— DAO Pattern

|— 5. SPRING FRAMEWORK

- | |
- | |— Spring Core
- | |— IoC
- | |— Dependency Injection
- | |— Beans
- | |— ApplicationContext
- | |— Component Scanning
- | |— @Component
- | |— @Service

- | |— @Repository
- | |— @Controller
- | — Constructor Injection
- |
- |— 6. SPRING BOOT ★
- | |
- | |— Spring Boot Setup
- | |— Project Structure
- | |— Starters
- | |— Auto Configuration
- | |— application.properties / yml
- | |— REST Controllers
- | |— @GetMapping
- | |— @PostMapping
- | |— @PutMapping
- | |— @DeleteMapping
- | |— @RequestBody
- | |— @PathVariable
- | |— @RequestParam
- | |— DTOs
- | |— Validation
- | |— Exception Handling
- | |— @ControllerAdvice
- | |— Profiles
- | — Actuator
- |
- |— 7. TESTING — INTRO
- | |
- | — JUnit Basics
- | |— @Test

- | └─ Assertions
- | └─ @BeforeEach
- |
- | └─ 8. JPA + HIBERNATE
- | |
- | └─ ORM
- | └─ Entity
- | └─ @Entity
- | └─ @Id
- | └─ @GeneratedValue
- | └─ JpaRepository
- | └─ CrudRepository
- | └─ One-to-One
- | └─ One-to-Many
- | └─ Many-to-One
- | └─ Many-to-Many
- | └─ JPQL
- | └─ @Query
- | └─ Transactions
- | └─ Lazy / Eager Loading
- | └─ Dirty Checking
- | └─ Pagination
- | └─ Sorting
- | └─ N+1 Problem
- |
- | └─ 9. SPRING SECURITY 🔒
- | |
- | └─ Authentication
- | └─ Authorization
- | └─ Password Hashing

- | |— Roles
- | |— Authorities
- | |— Security Filter Chain
- | |— JWT
- | |— Access Token
- | |— Refresh Token
- | |— CORS
- | |— CSRF
- | |— OAuth2 Basics

|

|— 10. TESTING — ADVANCED

| |

- | |— JUnit Advanced
- | |— Mockito
- | |— @Mock
- | |— @InjectMocks
- | |— MockMvc
- | |— Integration Testing
- | |— @SpringBootTest

|

|— 11. PRODUCTION BACKEND ★

| |

- | |— Logging
 - | | |— SLF4J
 - | | |— Logback
- | |— API Documentation
 - | | |— Swagger / OpenAPI
- | |— Redis
- | |— Caching
- | |— Cache Invalidation

- | |— Kafka
- | | |— Producer
- | | |— Consumer
- | | |— Topic
- | | |— Partition
- | | |— Offset
- | | |— Consumer Groups
- | |— Monitoring
- | | |— Actuator
- | | |— Metrics
- | | |— Health Checks
- | |— Distributed Tracing Basics

|

|— 12. MICROSERVICES ★

- | |
- | |— Monolith vs Microservices
- | |— Service Boundaries
- | |— API Gateway
- | |— Service Discovery
- | |— Eureka
- | |— OpenFeign
- | |— Config Server
- | |— Inter-Service Communication
- | |— Resilience4j
- | |— Retry
- | |— Timeout
- | |— Circuit Breaker
- | |— Distributed Transactions
- | |— Saga Pattern
- | |— Idempotency

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| — 13. DEVOPS 🌟

| |

| | — Linux Basics

| | — SSH

| | — Docker

| | | — Images

| | | — Containers

| | | — Dockerfile

| | | — Volumes

| | | — Networks

| | | — Environment Variables

| | — Docker Compose

| | — CI/CD

| | | — GitHub Actions

| | | — Jenkins

| | — Build

| | — Test

| | — Package

| | — Deploy

|

| — 14. CLOUD ☁️

| |

| | — AWS

| | | — IAM

| | | — EC2

| | | — S3

| | | — RDS

| | | — VPC Basics

| | | — Load Balancer

- | └─ Auto Scaling
- | └─ CloudWatch
- |
- └─ 15. KUBERNETES
- | |
- | └─ Containers vs Pods
- | └─ Pods
- | └─ Deployments
- | └─ Services
- | └─ ConfigMaps
- | └─ Secrets
- | └─ Namespaces
- | └─ Ingress
- | └─ Scaling
- |
- └─ 16. ADVANCED / PRO
- |
- └─ SOLID Principles
- |
- └─ Design Patterns
- | └─ Singleton
- | └─ Factory
- | └─ Builder
- | └─ Strategy
- | └─ Observer
- | └─ Adapter
- | └─ Decorator
- | └─ Facade
- | └─ Proxy
- |

└─ LLD

| └─ Class Design

| └─ Object Modeling

| └─ UML

|

└─ HLD

| └─ Scalability

| └─ Availability

| └─ Reliability

| └─ Load Balancing

| └─ Caching

| └─ Message Queues

| └─ Database Replication

| └─ Database Sharding

| └─ CAP Theorem

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└─ SYSTEM DESIGN

└─ URL Shortener

└─ Rate Limiter

└─ Notification System

└─ E-Commerce System

└─ Food Delivery System